Option E: Trend / Thematic Analysis

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Date:-29/07/2026

Title:-Abuse of rundll32.exe as a LOLBin

1.Trend Overview

The abuse of rundll32.exe as a Living-off-the-Land Binary (LOLBin) is a growing cybersecurity trend. Attackers use this legitimate Windows utility to execute malicious DLL files while appearing as a trusted system process. This technique helps them evade detection and is commonly used in malware, ransomware, and other cyberattacks.

2.Why Now?

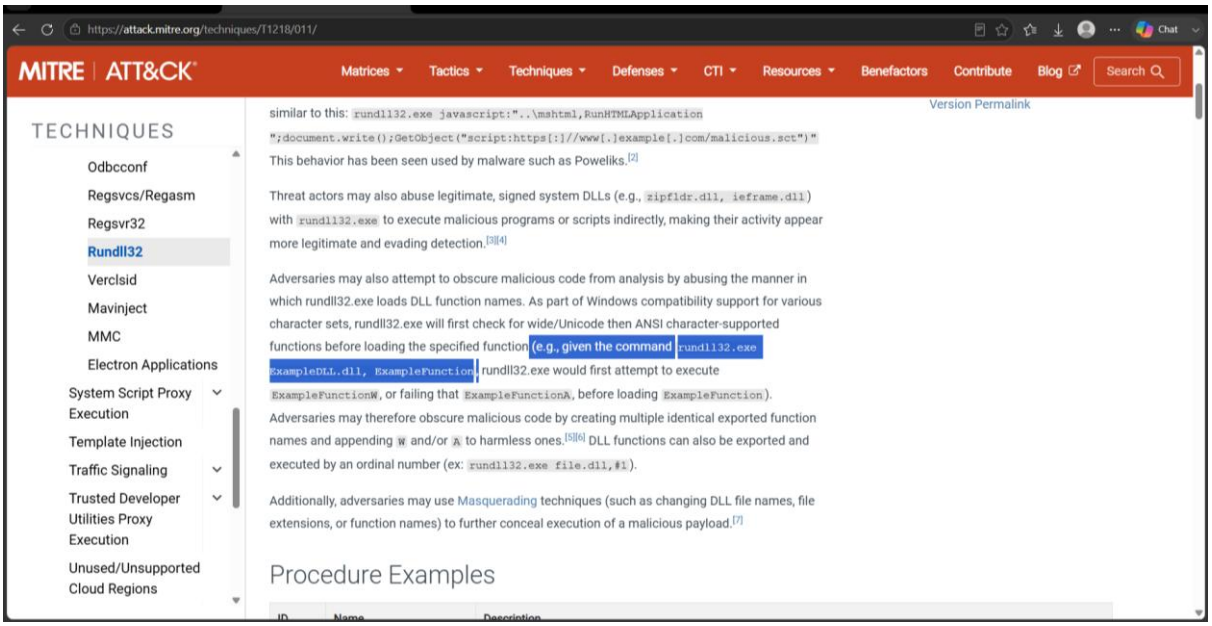
The abuse of rundll32.exe is increasing because it is a legitimate Microsoft-signed Windows utility that already exists on most systems. Attackers use it to execute malicious DLL files without dropping a new executable, making their activity appear legitimate. Since many security tools trust Windows system binaries, attackers can evade detection more easily by abusing "rundll32.exe".

3. Technical Details:

"rundll32.exe" is a legitimate Windows utility that executes functions stored in Dynamic Link Library (DLL) files. Attackers abuse this trusted binary by loading malicious DLLs

instead of legitimate ones. Because "rundll32.exe" is a Microsoft-signed process, its activity often appears normal and may bypass basic security controls. This technique allows attackers to execute malware, maintain persistence, and evade detection without using a suspicious executable file.

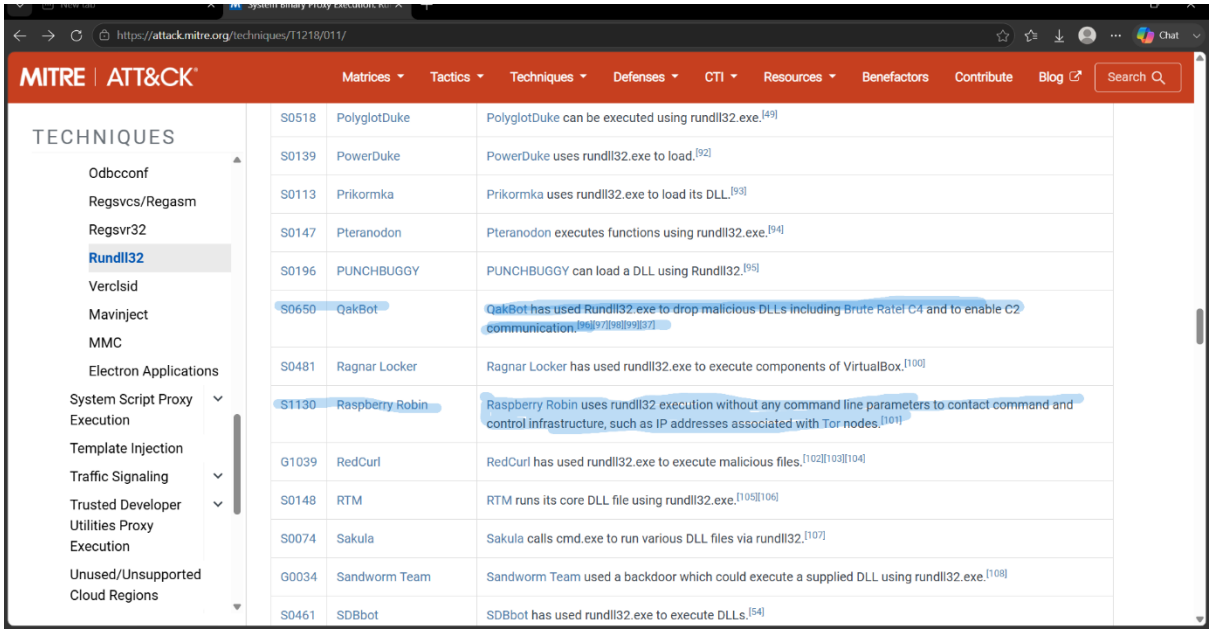
Example Command: rundll32.exe malicious.dll



4.Real-World Examples

1. QakBot (QBot): QakBot has abused "rundll32.exe" to execute malicious DLL files and establish command-and-control (C2) communication on infected systems.

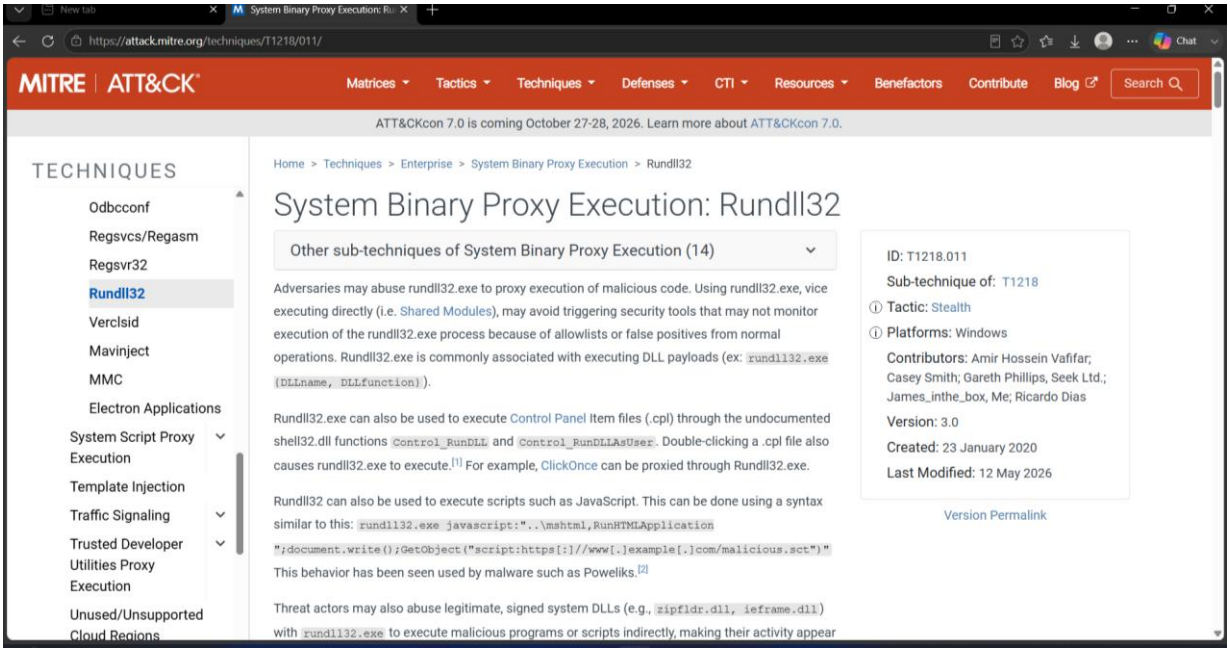
2. Raspberry Robin: Raspberry Robin uses "rundll32.exe" as a LOLBin to execute malicious code and communicate with its command-and-control (C2) infrastructure while evading detection.



5.MITRE ATT&CK Mapping

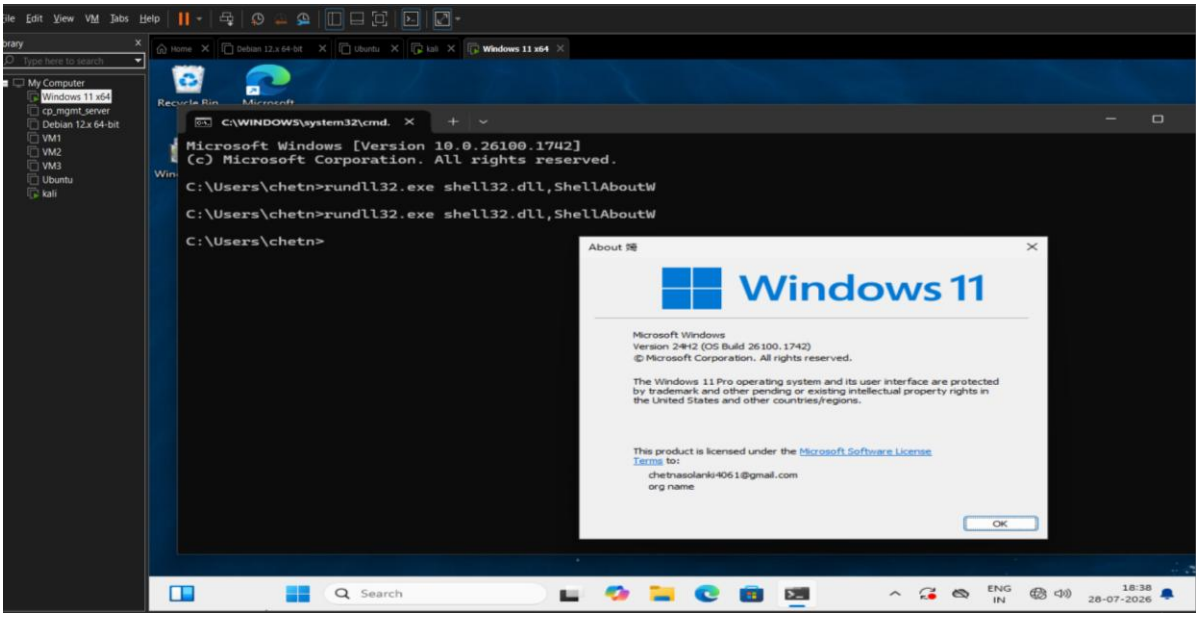
The abuse of "rundll32.exe" is mapped to the following MITRE ATT&CK technique:

- Technique ID: T1218.011
- Technique Name: Signed Binary Proxy Execution: Rundll32
- Description: Attackers abuse the legitimate "rundll32.exe" utility to execute malicious DLL files while avoiding detection.



6. Screenshot

Demonstration of rundll32.exe executing a legitimate DLL (shell32.dll) using the ShellAboutW function. This shows the normal behavior of rundll32.exe, which attackers abuse as a Living-off-the-Land Binary (LOLBin) to execute malicious DLLs.



7.Detection Opportunities

- Monitor suspicious rundll32.exe process execution.

- Detect rundll32.exe launching DLL files from unusual locations (e.g., Downloads, Temp, AppData).
- Use Sysmon Event ID 1 to monitor process creation.
- Deploy Sigma rules to detect suspicious rundll32.exe activity.
- Create Splunk or Elastic queries to identify abnormal command-line usage of rundll32.exe.

8.Defensive Recommendations

- Monitor and investigate suspicious rundll32.exe activity.
- Enable Microsoft Defender and EDR solutions for endpoint monitoring.
- Use AppLocker or Windows Defender Application Control (WDAC) to restrict unauthorized DLL execution.
- Keep Windows and security software up to date.
- Enable Sysmon and centralized logging to detect suspicious process activity.
- Train employees to avoid opening suspicious email attachments and files.

9.References

1.Reserch of rundll32:-

https://dev.to/itsmegsg/abusing-lolbins-rundll32exe-lab-sysmon-detection-170h

2.MITRK ATTECK Details:-

https://attack.mitre.org/techniques/T1218/011/

3.Red Canary:-

https://redcanary.com/threat-detection-report/techniques/rundll32/